

Data Dig: DDoS LOIT Attack Analysis

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Introduction

Simulating SOC-style detection and response to a real-world attack pattern

Dataset: CICIDS2017 [Canadian Institute for Cybersecurity, UNB] — simulated network traffic including DDoS, brute-force, web attacks, infiltration, and botnet activity.

Focus: Friday, July 7, 2017 — DDoS LOIT attack [15:56–16:16]

Playbook: Incident Response Consortium — Denial of Service Playbook [7-stage NIST-based process: Prepare → Detect → Analyze → Contain → Eradicate → Recover → Post-Incident]

Objective: Investigate whether a DDoS incident occurred, analyze its scope and impact, and walk through an incident response lifecycle using Splunk and Catalyst.

Preparation

- Studying dataset context
- Studying network architecture
- Reviewing DDos LOIT attacks
 - note common Indicators of Compromise
 - note common tactics, techniques and procedures
- Research-based Investigative questions that will be used in the Detection phase when querying our dataset with the appropriate tools

Dataset Background Study

What is CICIDS2017?

- A dataset generated by researchers at the Canadian Institute for Cybersecurity [CIC].
- **simulated network testbed** [virtual platform testing protocols, hardware, and distributed systems under **artificial network conditions** — mimic real-world network traffic, packet delays, and node failures]
 - victim machines sit behind a **firewall** with **NAT** [network address translation]
 - targets of attacks are **standard enterprise infrastructure, web servers** and **workstations** across multiple OS versions.
- Data is available as full packet captures [PCAP] or pre-processed CSV files.
- Created due to lack of reliable datasets for accurate evolution of anomaly-based intrusion detection approaches
 - **Complete Network Topology**
 - **Complete Traffic**
 - victim network and attack network
 - **Labelled Dataset**
 - show benign/attack labels per day
 - **Simulates Complete Interaction**
 - within & b/w internal LANs + Internet communication
 - **Complete Capture**
 - all traffic captured and recorded
 - **All Common Available Protocols**
 - e.g. HTTP, HTTPS, FTP, SSH and email protocols.
 - **Attack Diversity**
 - Web based, Brute force, DoS, DDoS, Infiltration, Heart-bleed, Bot and Scan
 - **MetaData**
 - Completely explained the dataset which includes the time, attacks, flows and labels in the published paper.

Simulated Network Testbed Architecture

Victim and attacker networks information

- **Firewall:** 205.174.165.80, 172.16.0.1
- **DNS+ DC Server:** 192.168.10.3
- **Outsiders [Attackers network]**
 - Kali: 205.174.165.73
 - Win: 205.174.165.69, 70, 71
- **Insiders [Victim network]**
 - Web server 16 Public: 192.168.10.50, 205.174.165.68
 - Ubuntu server 12 Public: 192.168.10.51, 205.174.165.66
 - Ubuntu 14.4, 32B: 192.168.10.19
 - Ubuntu 14.4, 64B: 192.168.10.17
 - Ubuntu 16.4, 32B: 192.168.10.16
 - Ubuntu 16.4, 64B: 192.168.10.12
 - Win 7 Pro, 64B: 192.168.10.9
 - Win 8.1, 64B: 192.168.10.5
 - Win Vista, 64B: 192.168.10.8
 - Win 10, pro 32B: 192.168.10.14
 - Win 10, 64B: 192.168.10.15
 - MAC: 192.168.10.25

Project's Scope of Analysis

- **Dataset Files:** MachineLearningCSV.zip and GeneratedLabelledFlows.zip
- **Date-Time:** Friday, July 7, 2017 – Afternoon
- **Type:** DDoS LOIT [15:56 – 16:16]
- **Attackers:** Three Win 8.1, 205.174.165.69 - 71
- **Victim:** Ubuntu16, 205.174.165.68 [Local IP: 192.168.10.50]
- **NAT Process on Firewall:**
 - Attackers: 205.174.165.69, 70, 71 -> 205.174.165.80 [Valid IP of the Firewall] -> 172.16.0.1

Reviewing DDoS LOIT Attacks

DDoS: [DDoS Attacks](#) [DDoS Mitigation](#)

- Launched from multiple connected online devices (botnet) distributed across the Internet to overwhelm target network infrastructure with huge volumes of traffic
- **Volume Based Attacks**
 - UDP floods, ICMP floods, other spoofed-packet floods.
 - saturates bandwidth of attacked site
 - magnitude in bits per second (Bps).
- **Protocol Attacks**
 - SYN floods, fragmented packet attacks, Ping of Death, Smurf DDoS, etc.
 - consumes resources of servers/intermediate communication equipment, e.g. firewalls, load balancers.
 - measured in packets per second (Pps).
- **Application Layer Attacks**
 - low-and-slow attacks, GET/POST floods, attacks that target Apache, Windows, or OpenBSD vulnerabilities, and more.
 - seemingly legitimate and innocent requests
 - aim to crash the web server
 - magnitude is measured in Requests per second (Rps).

LOIT: [LOIT Attacks](#) [LOIT Mitigation](#)

- popular open-source network stress testing & DoS tool
- often part of coordinated Distributed Denial of Service (DDoS) attacks.
- **TCP Flood**
 - Action: establishes multiple half-open connections that consume resources on the target server waiting to complete the handshake.
 - Consequence: This can overwhelm networks and load balancers.
- **UDP Flood**
 - Action: sends a stream of UDP packets to random destination ports on the target.
 - Consequence: saturates the network capacity.
- **HTTP Flood**
 - Action: sends continual HTTP GET or POST requests to specific URLs on the target website.
 - Consequence: This can consume web server resources or bandwidth.

Reviewing DDos LOIT Attacks

Indicator of Compromise/Attack/Impact	Relevance
Sudden bandwidth spike	Strong evidence of a volumetric DDoS
High PPS (Packets Per Second)	Strong evidence of a protocol-level flood
High RPS (Requests Per Second)	Strong evidence of an application-layer flood
Large number of distributed source IPs	Supports the distributed aspect of DDoS
Repetitive HTTP GET/POST requests	Strong indicator of HTTP flooding/LOIC HTTP mode
Large number of SYN/half-open connections	Strong indicator of TCP/SYN flooding
Large volume of UDP traffic/random destination ports	Particularly consistent with LOIC UDP flooding
Malformed/anomalous packets	Supports protocol-level DDoS
LOIC process/executable on an attacking host	Direct host-level evidence of LOIC involvement
IRC/C&C activity associated with attacking hosts	Supports coordination of multiple attacking machines
Known malicious IP reputation	Supports attribution of traffic to previously identified attack infrastructure
Multiple LOIC instances targeting the same destination	Directly connects LOIC activity with a distributed attack
Suspicious traffic spikes before the main attack	Potential evidence of reconnaissance or defensive "dry runs"

Research-based Investigative Questions

Question 1 - Did traffic suddenly increase?

Question 2 - When did it begin?

Question 3 - Who generated the traffic?

Question 4 - Who received the traffic?

Question 5 - Was one protocol abused?

Question 6 - Were other systems involved?

Question 7 - Did traffic return to normal?

Question 8 - Does the data match LOIC behavior?

Detection

- Importing dataset files
- Conducting searches on Splunk & Catalyst based on investigative questions

Detection: Splunk Searches We Ran

- **Attack window:** `index=cyb102_ddos Label="DDoS" | stats earliest[Timestamp] as First, latest[Timestamp] as Last, count as Total`
- **Port 80 breakdown:** `index=cyb102_ddos "Destination Port"="80" | stats count by Label`
- **Source/destination confirmation:** `index=cyb102_ddos Label="DDoS" | table Timestamp, "Source IP", "Destination IP", "Destination Port"`
- **Flow volume over time:** `index=cyb102_ddos Label="DDoS" | stats count by Timestamp | sort Timestamp`
- **Flow behavior comparison:** `index=cyb102_ddos | stats avg("Total Fwd Packets"), avg("Total Length of Fwd Packets"), avg("Flow Duration") by Label`



Analysis

- Categories of Investigation:

- Timeline
- Source Analysis
- Destination Analysis
- Protocol Analysis
- Port Analysis
- Flow Behavior

- DDoS attack takes place on Friday 07/07/2017 from 15:56 to 16:16.
- Largest peaks happen at 15:57 and 16:13.
- Attack ends at 16:16 and business operations start to continue normally



Events (128,027)

Patterns

Statistics (2)

Visualization

100 Per Page ▾

✎ Format

Preview ▾

Source IP ▾	✎	Destination IP ▾	✎	Protocol ▾	✎	total_events ▾	✎	unique_connections ▾	✎	unique_dest_ports ▾	✎
172.16.0.1		192.168.10.50		6		128024		45392		1	
192.168.10.50		172.16.0.1		6		3		1		3	

172.16.0.1
[Attacker
NAT-translated
address].

128,024
related
events.

45,393 unique
source ports.

One unique
destination
port - port 80.

Port analysis

This was a single-service DDoS, not a multi-vector attack. The attacker's LOIT tool concentrated entirely on overwhelming the web server's HTTP service.

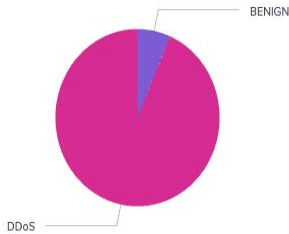
Most targeted destination port: Port 80 [HTTP]; Of 128,027 DDoS-labeled flows, **128,024 targeted port 80**. Only 3 flows hit other ports [64869, 64873, 27636 — one flow each], which are most likely noise/outliers, not a real secondary target.



Consistent with a targeted denial-of-service against one specific application rather than broad network disruption.

Ddos Attack Friday Afternoon

Port 80 Traffic — BENIGN vs. DDoS (Friday, July 7)



First_DDoS_Event ↕	Last_DDoS_Event ↕	Total_DDoS_Flows ↕
7/7/2017 4:04	7/7/2017 4:14	128027

DDoS Attack Table — Source/Destination IPs

Timestamp ↕	Source IP ↕	Destination IP ↕	Destination Port ↕
7/7/2017 4:14	172.16.0.1	192.168.10.50	80
7/7/2017 4:14	172.16.0.1	192.168.10.50	80
7/7/2017 4:14	172.16.0.1	192.168.10.50	80
7/7/2017 4:14	172.16.0.1	192.168.10.50	80
7/7/2017 4:14	172.16.0.1	192.168.10.50	80
7/7/2017 4:14	172.16.0.1	192.168.10.50	80
7/7/2017 4:14	172.16.0.1	192.168.10.50	80
7/7/2017 4:14	172.16.0.1	192.168.10.50	80
7/7/2017 4:14	172.16.0.1	192.168.10.50	80
7/7/2017 4:14	172.16.0.1	192.168.10.50	80
7/7/2017 4:14	172.16.0.1	192.168.10.50	80

DDoS Flow Count by Minute (Friday 3:56–4:16 PM)



Validating our Hypotheses

Traffic spike

Hypothesis: Traffic spikes 15:56–16:16, then returns to normal

Evidence: Flow chart shows sudden jump from 2,438 → 8,244 flows/min at 3:56 PM, sustained plateau, drop-off ~4:15 PM

SUPPORTED

Attacker IPs

Hypothesis: Three attacker IPs show high connection counts to victim

Evidence: All DDoS flows show single source IP [172.16.0.1] — dataset docs confirm NAT translates 3 attacker IPs → 1 internal IP

SUPPORTED

Resource Exhaustion

Hypothesis: Incomplete connections/retransmissions indicate resource exhaustion

Evidence: DDoS flows had *lower* avg bytes/packets per flow than benign traffic — exhaustion came from flow *volume*, not per-flow intensity

NOT SUPPORTED

Catalyst

Incidents

CAQL status == 'open' AND (owner == 'admin' OR

Name	Status	Owner	Creation	Last Modification
Incident #9468: IR-2023-002 Malware incident	Open	admin	2026-07-14 12:02:42	

Rows per page: 10 1-1 of 1

Incident #14128: DDoS LOIT Attack — Friday, July 7, 2017 (3:56–4:16 PM)

Open 2026-08-02 06:58:18 2026-08-02 07:04:42

Details

Severity High TLP White

Description

Simulated DDoS incident from the CICIDS2017 dataset (Friday, July 7, 2017), analyzed via Splunk as part of CYB102 capstone Milestone 2.

IOCs:

- Post-NAT source IP: 172.16.0.1

- Original attacker IPs: 205.174.165.69–71

- Victim: 192.168.10.50 (Ubuntu16 web server, public IP 205.174.165.68)

- Destination port: 80 (HTTP)

Key findings:

- 128,027 DDoS-labeled flows vs. 8,927 benign flows on port 80 (~14:1 ratio)

- Sustained flood pattern: 6,000–8,000 flows/minute from 3:56–4:16 PM

- Attack traffic NAT-translated to a single internal IP before reaching victim

Playbook reference: Incident Response Consortium — Denial of Service Playbook

Owner admin

Playbooks Simple

References

Artifacts

Related Tickets

Files

CLOSE

HANDBOOK

SAVE DETAILS

Answering Investigative Questions

Question 1 - Did traffic suddenly increase?

Yes there was a sudden jump from 2k flows to 8k flows right at the attack start [3:56-3:57]

Question 2 - When did it begin?

3:56pm, July 7, 2017

Question 3 - Who generated the traffic?

Source IP 172.16.0.1[NAT-translated from three attacker IPs: 205.174.165.69–71]

Question 4 - Who received the traffic?

192.168.10.50 [Ubuntu16 web server]

Question 5 - Was one protocol abused?

HTTP - all traffic targeted port 80

Question 6 - Were other systems involved?

No — 192.168.10.50 was the only destination across all 128,027 DDoS flows

Question 7 - Did traffic return to normal?

Yes — flow counts dropped sharply after 4:15–4:16 PM

Question 8 - Does the data match LOIC behavior?

Partially — high volume/few sources matched, but per-flow byte/packet rates were actually lower than benign traffic

Impact

- Possibility of degrading services
 - Abnormal amounts of connections can impact availability of services
- Severity: Medium
 - Confidentiality/Integrity impact is low if not none.
 - Availability is high risk, sustained DDoS is capable or degrading or denying access to services.

Response

Containment



- Rate limiting
 - Cap new connections per second per source IP
- Blocking malicious IPs
 - Block 172.16.0.1, the confirmed source of all 128,000+ attack events
- Configuring a Web Application Firewall (WAF)
 - Filter high-volume repeated requests to port 80 before they reach the server

Response

Eradication



- Remove malicious traffic
 - Terminate any connections from 172.16.0.1 that are still open in the server's connection table
- Confirm persistence does not exist
 - Verify no backdoor or secondary compromise occurred during the attack

Response

Recovery



- Restore normal routing
 - Traffic returned to baseline within 1 minute of the attack ending
- Monitor service performance
 - Check CPU, memory, and connection table on 192.168.10.50
- Validate service availability
 - Confirm port 80 responds normally to real user traffic
- Make containment measures permanent
 - Formalize the temporary block/rate-limit rules into standing policy

Post-Incident Handling

Lessons Learned, Improvements, Future Recommendations

- Single IP was able to send abnormal amount of connections to a single port.
 - No mitigation methods were present to stop it
- Establish a baseline of "normal" traffic volume/connection patterns

Thank you!

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